

Triggers Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To create and test triggers that fire automatically BEFORE or AFTER an INSERT, UPDATE, or DELETE.

A trigger needs a log/audit table to show its effect. We create a small FACULTY_LOG table below to prove each trigger actually ran.

-- 0. Support table: log of changes to FACULTY

```
CREATE TABLE FACULTY_LOG (  
    LogID    INT AUTO_INCREMENT PRIMARY KEY,  
    Action   VARCHAR(10),  
    FacNo    VARCHAR(4),  
    ActionTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
mysql> CREATE TABLE FACULTY_LOG (  
->     LogID    INT AUTO_INCREMENT PRIMARY KEY,  
->     Action   VARCHAR(10),  
->     FacNo    VARCHAR(4),  
->     ActionTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

-- 1. AFTER INSERT trigger

```
DELIMITER //  
CREATE TRIGGER trg_after_insert_faculty  
AFTER INSERT ON FACULTY  
FOR EACH ROW  
BEGIN  
    INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('INSERT', NEW.FacNo);  
END //  
DELIMITER ;  
  
INSERT INTO FACULTY (FacNo, FacName, Gender, DOB, DOJ, MobileNo, Dept, Age)  
VALUES ('F006', 'Priya', 'F', '1985-04-10', '2012-03-01', 9840099999, 'IT', 41);  
  
SELECT * FROM FACULTY_LOG;
```

```

mysql> DELIMITER //
mysql> CREATE TRIGGER trg_after_insert_faculty
-> AFTER INSERT ON FACULTY
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('INSERT', NEW.FacNo);
-> END //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER ;
mysql>
mysql> INSERT INTO FACULTY (FacNo, FacName, Gender, DOB, DOJ, MobileNo, Dept, Age)
-> VALUES ('F006', 'Priya', 'F', '1985-04-10', '2012-03-01', 9840099999, 'IT', 41);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM FACULTY_LOG;
+-----+-----+-----+-----+
| LogID | Action | FacNo | ActionTime          |
+-----+-----+-----+-----+
|      1 | INSERT | F006  | 2026-08-13 14:16:23 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

```

-- 2. BEFORE UPDATE trigger (validate/adjust incoming data)

```

DELIMITER //
CREATE TRIGGER trg_before_update_faculty
BEFORE UPDATE ON FACULTY
FOR EACH ROW
BEGIN
    IF NEW.Age < 0 THEN
        SET NEW.Age = 0;
    END IF;
END //
DELIMITER ;

UPDATE FACULTY SET Age = -5 WHERE FacNo = 'F006';
SELECT FacNo, Age FROM FACULTY WHERE FacNo = 'F006';    -- Age should show 0, not -5

```

```

mysql> DELIMITER ;
mysql>
mysql> UPDATE FACULTY SET Age = -5 WHERE FacNo = 'F006';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT FacNo, Age FROM FACULTY WHERE FacNo = 'F006';    -- Age should show 0, not -5
+-----+-----+
| FacNo | Age |
+-----+-----+
| F006  | 0   |
+-----+-----+
1 row in set (0.00 sec)

```

-- 3. AFTER UPDATE trigger (log the change)

```
DELIMITER //
CREATE TRIGGER trg_after_update_faculty
AFTER UPDATE ON FACULTY
FOR EACH ROW
BEGIN
    INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('UPDATE', NEW.FacNo);
END //
DELIMITER ;
```

```
UPDATE FACULTY SET Dept = 'CSE' WHERE FacNo = 'F006';
SELECT * FROM FACULTY_LOG WHERE FacNo = 'F006';
```

```
mysql> DELIMITER ;
mysql>
mysql> UPDATE FACULTY SET Dept = 'CSE' WHERE FacNo = 'F006';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM FACULTY_LOG WHERE FacNo = 'F006';
+-----+-----+-----+-----+
| LogID | Action | FacNo | ActionTime          |
+-----+-----+-----+-----+
|      1 | INSERT | F006  | 2026-08-13 14:16:23 |
|      2 | UPDATE | F006  | 2026-08-13 14:17:21 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

-- 4. BEFORE DELETE trigger (prevent deleting a department head)

```
DELIMITER //
CREATE TRIGGER trg_before_delete_faculty
BEFORE DELETE ON FACULTY
FOR EACH ROW
BEGIN
    IF EXISTS (SELECT 1 FROM DEPT WHERE DeptHead = OLD.FacNo) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Cannot delete a faculty member who heads a department';
    END IF;
END //
DELIMITER ;
```

```
-- This should now fail with a custom error message:
DELETE FROM FACULTY WHERE FacNo = 'F001';
```

```

mysql> DELIMITER //
mysql> CREATE TRIGGER trg_before_delete_faculty
-> BEFORE DELETE ON FACULTY
-> FOR EACH ROW
-> BEGIN
->     IF EXISTS (SELECT 1 FROM DEPT WHERE DeptHead = OLD.FacNo) THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Cannot delete a faculty member who heads a department';
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> -- This should now fail with a custom error message:
mysql> DELETE FROM FACULTY WHERE FacNo = 'F001';
ERROR 1644 (45000): Cannot delete a faculty member who heads a department

```

-- 5. AFTER DELETE trigger (log the deletion)

```

DELIMITER //
CREATE TRIGGER trg_after_delete_faculty
AFTER DELETE ON FACULTY
FOR EACH ROW
BEGIN
    INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('DELETE', OLD.FacNo);
END //
DELIMITER ;

DELETE FROM FACULTY WHERE FacNo = 'F006'; -- not a dept head, so this succeeds
SELECT * FROM FACULTY_LOG WHERE FacNo = 'F006';

```

```

mysql> DELIMITER ;
mysql>
mysql> DELETE FROM FACULTY WHERE FacNo = 'F006'; -- not a dept head, so this succeeds
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM FACULTY_LOG WHERE FacNo = 'F006';
+-----+-----+-----+-----+
| LogID | Action | FacNo | ActionTime |
+-----+-----+-----+-----+
| 1 | INSERT | F006 | 2026-08-13 14:16:23 |
| 2 | UPDATE | F006 | 2026-08-13 14:17:21 |
| 3 | DELETE | F006 | 2026-08-13 14:18:17 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

-- 6. Viewing and dropping triggers

```

SHOW TRIGGERS FROM lab;
SHOW CREATE TRIGGER trg_after_insert_faculty;

```

DROP TRIGGER IF EXISTS trg_after_insert_faculty;

```
-----+-----+-----+-----+-----+
| trg_after_insert_faculty | INSERT | faculty | BEGIN
| INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('INSERT', NEW.FacNo);
END
ER | 2026-08-13 14:16:23.78 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_before_update_faculty | UPDATE | faculty | BEGIN
| IF NEW.Age < 0 THEN
| SET NEW.Age = 0;
| END IF;
END
| BEFORE | 2026-08-13 14:16:49.81 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_after_update_faculty | UPDATE | faculty | BEGIN
| INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('UPDATE', NEW.FacNo);
END
ER | 2026-08-13 14:17:21.30 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_before_delete_faculty | DELETE | faculty | BEGIN
| IF EXISTS (SELECT 1 FROM DEPT WHERE DeptHead = OLD.FacNo) THEN
| SIGNAL SQLSTATE '45000'
| SET MESSAGE_TEXT = 'Cannot delete a faculty member who heads a department';
| END IF;
END
| BEFORE | 2026-08-13 14:17:43.17 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_after_delete_faculty | DELETE | faculty | BEGIN
| INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('DELETE', OLD.FacNo);
END
ER | 2026-08-13 14:18:17.42 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> SHOW CREATE TRIGGER trg_after_insert_faculty;
-----+-----+-----+-----+
| Trigger | sql_mode | SQL Original Statement | character_set_client | collation_connection | Database |
-----+-----+-----+-----+
| trg_after_insert_faculty | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | CREATE DEFINER='root'@'localhost' TRIGGER 'trg_after_insert_faculty' AFTER INSERT ON 'faculty' FOR EACH ROW BEGIN
| INSERT INTO FACULTY_LOG(Action, FacNo) VALUES ('INSERT', NEW.FacNo);
END | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci | 2026-08-13 14:16:23.78 |
-----+-----+-----+-----+
1 row in set (0.02 sec)

mysql> DROP TRIGGER IF EXISTS trg_after_insert_faculty;
Query OK, 0 rows affected (0.01 sec)
```